SSVEP EEG Data Collection Protocol Using EMOTIV EPOC X

1. Objective

To collect Steady-State Visual Evoked Potential (SSVEP) EEG recordings using the EMOTIV EPOC X headset while participants attend to visual stimuli flickering at 10 Hz and 12 Hz.

Two visual stimuli shall be used:

- Red stimulus box flickering at 10 Hz
- Blue stimulus box flickering at 12 Hz

The dataset will be used for:

- EEG preprocessing
- Frequency-domain analysis
- SSVEP characterization
- Brain-Computer Interface (BCI) research
- Attention-based SSVEP classification

2. Equipment

EEG Acquisition

- EMOTIV EPOC X (14-channel EEG headset)
- EMOTIV USB Dongle
- EmotivPRO Software
- Saline solution

Sampling Rate:

- 128 Hz acceptable

Stimulus Presentation

- Laptop/Desktop Computer
- 60 Hz Monitor
- Python
- Pygame

Data Storage

- CSV Export Format
- Event Timestamp Log

3. EEG Channels of Interest

Recorded Channels:

AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, AF4

Primary Channels

- O1
- O2

Secondary Channels

- P7
- P8

All channels shall be recorded, with O1 and O2 serving as the primary channels for SSVEP analysis.

4. Participant Criteria

Inclusion Criteria

- Age 18–35 years
- Normal or corrected-to-normal vision
- Able to understand instructions
- Minimum 6 hours of sleep before recording

Exclusion Criteria

- Photosensitive epilepsy
- Neurological disorders
- Severe visual impairment
- Excessive fatigue
- CNS-active medication

Safety Screening

Participants shall be asked:

“Have you ever experienced a seizure triggered by flashing lights or flickering images?”

Participants answering **Yes** or **Unsure** shall not participate.

5. Experimental Environment

Parameter	Requirement
Lighting	Dark Room setting
Monitor Refresh Rate	60 Hz
Screen Brightness	70–80%
Viewing Distance	60 cm
Room Noise	Minimal
Temperature	20–25°C

Participants shall remain comfortably seated throughout the session.

6. Participant Preparation

Participants shall:

- Obtain at least 6 hours of sleep
- Avoid caffeine for 2 hours before recording
- Avoid hair oil, gel, or styling products
- Remove metallic accessories near the head

7. Headset Setup

Sensor Preparation

- Hydrate all sensors using saline solution
- Ensure O1, O2, CMS, and DRL sensors are adequately wetted

Contact Quality Check

Verify channel quality using EmotivPRO.

Recording shall begin only when:

- O1 quality is acceptable
- O2 quality is acceptable
- CMS quality is acceptable
- DRL quality is acceptable

8. Signal Verification Tests

Noise Check

Duration: 30 seconds

Expected Result:

- Stable EEG signal
- Minimal interference

Alpha Verification

- Eyes Closed: 10 seconds
- Eyes Open: 10 seconds

Expected Result:

- Increased alpha activity (8–13 Hz) at O1 and O2

Blink Test

- Blink slowly five times

Expected Result:

- Visible blink artefacts in frontal channels

9. Visual Stimulus Design

Stimulus Frequencies

Stimulus	Frequency
Red Box	10 Hz
Blue Box	12 Hz

Visual Layout

The stimulus screen shall consist of:

- Grey background
- Left red stimulus box
- Right blue stimulus box

Baseline and Rest State

During baseline and rest periods:

- Grey background with a crosshair is shown

No flickering stimulus shall be presented.

Red SSVEP Condition

- Left red box flickers at 10 Hz
- Flicker generated by alternating between:
    - Bright red
    - Dim red
- Right blue box remains dim

Blue SSVEP Condition

- Right blue box flickers at 12 Hz
- Flicker generated by alternating between:
    - Bright blue
    - Dim blue
- Left red box remains dim

Stimulus Parameters

- Fullscreen presentation
- Grey background throughout experiment

- Swapping the sides of Red and Blue after 10 stimulus to prevent habituality.
- 50% duty cycle
- Python + Pygame implementation
- Monitor refresh rate fixed at 60 Hz

Colour Specification

Element	RGB
Bright Red	(255, 0, 0)
Dim Red	(80, 0, 0)
Bright Blue	(0, 170, 255)
Dim Blue	(0, 60, 80)
Grey Background	(128, 128, 128)

10. Baseline Recording

Duration

60 seconds

Procedure

Participants shall:

- Keep eyes open
- Remain relaxed
- Minimize movement

Both stimulus boxes shall remain visible in their dim state.

No flickering stimulus shall be presented.

11. Practice Session

Duration

Approximately 40 seconds

Purpose

- Familiarize participants with stimuli
- Reduce learning effects

- Confirm participant understanding

Practice Sequence:

- Red stimulus (5 s)
- Rest (5 s)
- Blue stimulus (5 s)
- Rest (5 s)
- Red stimulus (5 s)
- Rest (5 s)
- Blue stimulus (5 s)
- Rest (5 s)

Practice data shall not be included in the final dataset.

12. Experimental Recording Procedure

Each trial consists of:

- 5-second stimulus presentation
- 5-second rest interval

Participant Instructions:

- Attend to the target stimulus
- Avoid head movement
- Remain attentive
- Minimize blinking during stimulus periods
- Blink during rest intervals if required

Red Trials

Participants shall attend to:

- Red stimulus box

Blue Trials

Participants shall attend to:

- Blue stimulus box

The non-target stimulus shall remain visible in a dim state.

13. Session Structure

Baseline

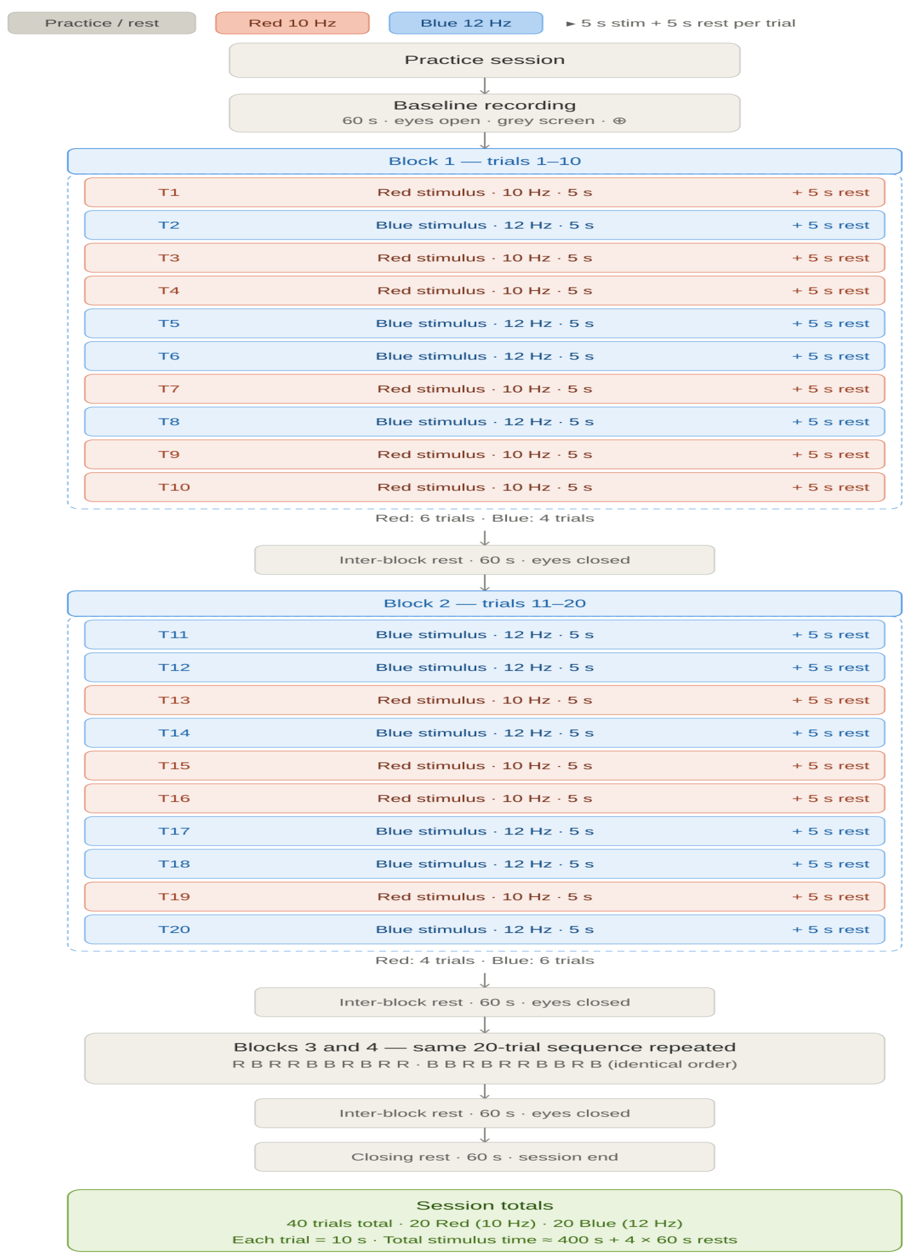
- 60 seconds

Practice Session

- 2 cycles

Cycle Structure:

- Red (5 s)
- Rest (5 s)
- Blue (5 s)
- Rest (5 s)



Dataset Yield

Per Participant:

- 20 trials at 10 Hz
- 20 trials at 12 Hz
- 40 labelled SSVEP trials

14. Event Logging

The following events shall be recorded:

- BASELINE_START
- BASELINE_END
- PRACTICE_START
- PRACTICE_END
- BLOCK_A_START
- BLOCK_A_END
- BLOCK_B_START
- BLOCK_B_END
- BLOCK_C_START
- BLOCK_C_END
- BLOCK_D_START
- BLOCK_D_END
- RED_START
- RED_END
- BLUE_START
- BLUE_END
- REST_START
- REST_END

All event timestamps shall be synchronized with EEG recordings.

15. Dataset Validation

Expected Spectral Responses

Red Trials

Expected Peaks:

- 10 Hz
- 20 Hz harmonic

Blue Trials

Expected Peaks:

- 12 Hz
- 24 Hz harmonic

Strongest responses are expected at:

- O1
- O2

Expected Dataset Contents

- 60-second baseline recording
- Practice recording
- 20 trials of 10 Hz SSVEP
- 20 trials of 12 Hz SSVEP
- Event timestamp log
- Session metadata
- Raw EEG recordings

16. References

1. Mao et al. (2017). Progress in EEG-Based Brain Robot Interaction Systems.
2. Asanza et al. (2021). SSVEP-EEG Signal Classification Based on Emotiv EPOC BCI and Raspberry Pi.
3. Nakanishi et al. (2015). Comparison of CCA-Based Methods for SSVEP Detection.
4. Radüntz (2020). Signal Quality Evaluation of Emerging EEG Devices.
5. Jasper (1958). The Ten-Twenty Electrode System.
6. Regan (1989). Human Brain Electrophysiology.